

1.7 Classwork: Financial calculations and compound interest — Markscheme

1. [Maximum mark: 6]

Maria invests \$2000 at 3.5% per year simple interest.

(a) $I = 2000 \times \frac{3.5}{100} = \70 **A1**

A value of \$7000 (using 3.5 in place of 0.035) earns no mark.

(b) \$2070, \$2140, \$2210 **A1**

All three values required for the mark.

FT: award A1 for $2000 + I$, $2000 + 2I$, $2000 + 3I$ with the candidate's own I from (a).

(c) $V_n = 2000 + 70n$ **A1**

Accept $V_n = 2070 + 70(n - 1)$ or any equivalent form.

$V_n = 2000 + 70(n - 1)$ earns no mark: it gives \$2000 at the end of year 1.

FT: award A1 for $2000 + In$ with the candidate's own I from (a).

(d) $V_{12} = 2000 + 70 \times 12 = \2840 **A1**

FT: award A1 for the value obtained by substituting $n = 12$ into the candidate's own expression from (c).

(e) $2000 + 70n > 3000$ **M1**

$V_{14} = \$2980$ (< 3000) and $V_{15} = \$3050$ (> 3000)

15 years **A1**

Accept solving the inequality: $n > 14.28\dots$, so 15 years.

Accept a table search: full marks for showing both adjacent years and stating 15. A single row (after 15 years the value is \$3050, so 15) still earns full marks, with the feedback note that the adjacent year 14 is what justifies “first”. See markscheme-spec §18.

14.3 is not an acceptable final answer — the answer must be a whole number of years. Rounding down to 14 earns the M1 only.

FT: award A1 for the smallest whole number of years correctly obtained from the candidate's own expression in (c).

2. [Maximum mark: 7]

\$2000 invested. Bank A: 3.5% compounded annually. Bank B: 3.4% compounded monthly.

$$(a) 2000 \times 1.035^{12} \quad \text{M1}$$

$$= \$3022.137 \dots \approx \$3022.14 \quad \text{A1}$$

Accept \$3020 (3 s.f.), \$3022 or \$3022.14. Accept the TVM solver: $N = 12$, $I\% = 3.5$, $PV = -2000$, $PMT = 0$, $P/Y = C/Y = 1$.

A multiplier of 0.035 or 1.35 earns no marks. An exponent of 11 (giving \$2919.94) earns the M1 only.

Simple-interest working ($2000 + 12 \times 70 = 2840$) earns no marks.

$$(b) 2000 \left(1 + \frac{3.4}{100 \times 12}\right)^{12 \times 12} \quad \text{M1}$$

$$= \$3005.879 \dots \approx \$3005.88 \quad \text{A1}$$

Accept \$3010 (3 s.f.), \$3006 or \$3005.88. Accept the TVM solver: $N = 144$, $I\% = 3.4$, $PV = -2000$, $PMT = 0$, $P/Y = C/Y = 12$.

M1 for a monthly rate of $\frac{3.4}{1200}$ (or 0.002833...) and an exponent of 144. Using 3.4% compounded annually (giving \$2987.28) earns no marks.

$$(c) \text{ Bank A} \quad \text{A1}$$

FT: award A1 for the bank with the greater value from the candidate's own answers to (a) and (b).

$$(d) 2000 \times 1.035^n \geq 4000 \quad \text{M1}$$

After 20 years: \$3979.58 (< 4000); after 21 years: \$4118.86 (≥ 4000)

$$21 \text{ years} \quad \text{A1}$$

Accept $1.035^n \geq 2$ solved on the GDC or with logarithms: $n \geq 20.15 \dots$, so 21 years. Accept the TVM solver for N with $PV = -2000$, $FV = 4000$.

Accept a table search: full marks for showing both adjacent years and stating 21. A single row (after 21 years the value is \$4118.86, so 21) still earns full marks, with the feedback note that the adjacent year 20 is what justifies "first". See markscheme-spec §18.

20.1 is not an acceptable final answer — the answer must be a whole number of years. Rounding down to 20 earns the M1 only.